

Amritanshu Kumar

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Professional Summary

Unity architect with **3.5 years** building scalable systems end to end — real-time multiplayer, physics simulation, performance-critical gameplay — and leading the teams that ship them. Directs a 7-engineer team on a live title with **41K registered / ~3.4K daily active** players. Built a custom C++ physics engine from scratch. **5M+ downloads** shipped. Deep hands-on experience with state machines, event-driven architecture, and debugging complex production issues under pressure.

Professional Experience

RootHoot Pvt Ltd

Team Lead

Remote, Hyderabad

Dec 2024 – Present

- Own the architecture across gameplay systems, networking, and API for Pongz, a live multiplayer Unity title shipping on Android and iOS.
- Architected the game's scalable Unity architecture around state machines, event-driven dispatch, and data-driven design. Replaced scattered boolean flags with a hierarchical state controller (8 phases, 14+ substates) — single source of truth for gameplay, UI, and networking.
- Engineered a server-authoritative multiplayer protocol that eliminated race conditions from latency and message reordering — monotonic turn sequencing, fingerprint-based dedup on reconnect, and exponential backoff with jitter.
- Built the connection layer with state-machine-enforced lifecycle management, typed event dispatch, offline queue with TTL expiry, and adaptive connectivity probing — optimized for stable real-time performance across variable network conditions.
- Designed a three-stage AFK system to handle player drops mid-match — warning, server-tracked idle, then AI-driven bot substitution. Full state resync on return without disrupting the other three players.
- Extracted 8 core Unity systems into shared team infrastructure — Event Bus, State Machine, Save, Pause, Logging, UI, Loading, Popup — with separated event/message buses, WeakReference lifecycle management, and atomic-write persistence.
- Directed architecture, code reviews, and sprint planning while staying hands-on in the codebase. Led features from design through implementation to delivery.

DeftSoft

Game Developer

Mohali, Punjab

Feb 2023 – Oct 2024

- Shipped a Unity title to **5M+ downloads** and **4.6 stars** on Google Play. Iterated on player data post-launch to push engagement up **30%**.
- Optimized CPU and rendering performance to hit **25%** higher FPS on low-end devices using Unity Profiler and Frame Debugger. Built procedural generation tools with data-driven design — cut level-design time by **40%**.
- Introduced TDD across the team to catch regressions before release — ratings improved **1.5 stars** as a direct result. Authored a **25-chapter** technical course on game architecture and design patterns.
- Mentored 3 junior developers — code reviews, pair programming, and coding standards.

Selected Projects

Nimirta Engine: Custom 2D physics engine, C++/SFML — impulse solver with warm starting, Baumgarte correction for stable high-velocity contacts, semi-implicit Euler, fixed timestep with spiral-of-death prevention, 3 collision algorithms, 8 material types, hand-written vector math, 3-tier AI system.

InstaReload: Unity hot-reload tool that cuts domain reload from 3–5s to **7ms**, removing the main bottleneck in the edit-test loop. Deep Unity internals: Roslyn for IL patching on debug edits (90% of cases), MonoMod detours for structural changes.

FlowUI: Open-source Unity UI framework (MIT) that eliminates UI boilerplate — CRTP builders, Roslyn codegen with partial classes, bidirectional O(1) lookups, and an 8-tab visual editor. Fully documented.

Technical Skills

- Languages & Engines:** C#, C++, TypeScript, Go, Python | Unity (primary), Unreal, Custom C++ Engine
- Unity Expertise:** Physics, State Machines, Event-Driven Architecture, Scriptable Objects, Prefab Architecture, Coroutines, Object Pooling, Profiler
- Patterns:** SOLID, State Machine, Observer, Factory, Strategy, ECS, Data-Driven Design
- Multiplayer:** Server-Authoritative Protocol, State Synchronization, Socket.IO, Reconnect/Resync
- Backend:** Node.js, Express, MongoDB, Redis, Docker, Google Cloud (App Engine), Firebase

Education & Certifications

Bachelor of Computer Application, Lovely Professional University, Punjab, India | Sep 2020 – Jun 2023

Certifications: Unity Certified Programmer | Unreal Engine Game Design & Development Specialization